

Derbyshire Shared Care Pathology Guidelines

Management of AKI detected in primary care

Recommended response times to AKI warning stage test results for adults in primary care:

AKI Warning Stage Test Result Confirm or refute automated AKI test result by comparing patient's current creatinine within clinical context against baseline creatinine	Clinical Context Within Which Blood Test Taken# If clinical context unknown, assume high pre-test probability until proven otherwise	
	LOW Pre-test probability of AKI Stable clinical context	HIGH Pre-test probability of AKI Context of acute illness
AKI1 Creatinine ≥ 1.5 x baseline (or creatinine rise >26 $\mu\text{mol/L}$ 48 hrs)	Consider clinical review ≤ 72 hrs of e-alert* If AKI confirmed → manage as per table on page 2	Consider clinical review ≤ 24 hrs of e-alert* Likely stage 1 AKI → manage as per table on page 2
AKI2 Creatinine ≥ 2 x baseline	Consider clinical review ≤ 24 hrs of e-alert* If AKI confirmed → manage as per table on page 2	Consider clinical review ≤ 6 hrs of e-alert* Likely stage 2 AKI → manage as per table on page 2
AKI3 Creatinine ≥ 3 x baseline (or creatinine 1.5 x baseline and >354 $\mu\text{mol/L}$)	Consider clinical review ≤ 6 hrs of e-alert* If AKI confirmed → consider admission	Consider immediate admission* Likely stage 3 AKI

#Clinical Context

Why was the blood test taken?

- Routine chronic disease monitoring
- Drug monitoring
- Assessment of acute illness

Creatinine rise within stable clinical context may reflect unstable CKD instead of AKI, especially if longer time period between current and baseline creatinine

*AKI Risk Factors / Clinical Features Prompting Earlier review

- Poor oral intake / urine output
- Evidence of hyperkalaemia, especially if moderate ($K^+ 6.0-6.4$) or severe ($K^+ \geq 6.5$)
- Known history of CKD stages 4 & 5 or history of kidney transplant
- Deficient immunity
- Frail with co-morbidities (CKD, diabetes, heart failure, liver disease, neurological or cognitive impairment)
- Past history of AKI
- Suspected intrinsic kidney disease
- Suspected urinary tract obstruction

Taken from [Table-1-Primary-Care-190117.pdf \(thinkkidneys.nhs.uk\)](https://www.thinkkidneys.nhs.uk/primary-care/190117)

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Management of adults with AKI detected in primary care:

"Think" Cause	"Think" Medication	"Think" Fluids	"Think" Review
History of acute illness? - Think sepsis - Think Hypotension Intrinsic Kidney Disease? (E.g. vasculitis) - Think Urinalysis Urinary Tract Obstruction?	Any medication which could exacerbate AKI? Consider withholding: - NSAIDs - Diuretics - Antihypertensive medication Any medication which may accumulate and cause harm during AKI? Any new medication that may cause AKI? (E.g. drug induced tubulo-interstitial nephritis)	What is the patient's volume status? If hypovolaemia present: - When did patient last pass urine? - Can the patient increase fluid intake? - Is admission for iv fluid replacement and monitoring required? Does the patient have and/or need carer support?	Does the patient need acute admission? If not, when will you review? Have you ensured handover?

Taken from [Recognising-and-Responding-to-Acute-Kidney-Injury-in-Primary-Care-Table-2.pdf \(thinkkidneys.nhs.uk\)](#)

Refer to background information on pages 3 to 8 as necessary or use hyperlinks below:

[What is AKI?](#)

[Why is AKI important?](#)

[Why is AKI important for primary care?](#)

[How to recognise AKI](#)

[What causes AKI?](#)

[Who is at risk of developing AKI?](#)

[How to reduce the risk of AKI](#)

[How to manage a patient with AKI detected in primary care](#)

[Appendix: AKI sick day rules leaflet](#)

[Other useful resources](#)

What is AKI?

Acute kidney injury (AKI) was previously known as acute kidney failure and simply means a sudden reduction in kidney function. It is **not** a traumatic injury to the kidney as the name may imply, nor is it a diagnosis in its own right, rather a syndrome with many different underlying causes.

Why is AKI important?

AKI is extremely common in hospitalised patients, occurring in 10-20% of emergency hospital admissions and is associated with extremely poor outcomes. However, AKI is not just a secondary care problem – primary care has a crucial role to play, particularly in prevention and post-AKI care (see below).

Poor outcomes associated with AKI:

- Extremely high mortality rates (more than 20% of patients with AKI will die during hospital admission, rising to >35% in those with AKI stage 3).
- Increased length of hospital stay and higher healthcare resource utilisation
- Failure of kidney recovery – episodes of AKI cause and increase progression of Chronic Kidney Disease (CKD)
- Increased risk of poor long term outcomes: life expectancy, cardiovascular risk, quality of life

In part, these poor outcomes reflect the fact that AKI acts as a 'force multiplier' and increases severity of any co-existing acute illness. **As such, AKI is a marker of the 'sick patient' who requires prompt recognition and management.**

It is also important that there are significant opportunities to improve AKI care. This is recognised in a major, national programme from NHS England called 'Think Kidneys' (<https://www.thinkkidneys.nhs.uk/>). This programme aims to reduce avoidable harm associated with AKI across all healthcare settings.

Why is AKI important for Primary Care?

- Up to two-thirds of patients who sustain AKI have already developed this by the time they are admitted to hospital, so preventative strategies have to include pre-hospital care.
- Automatic recognition of AKI based on outpatient bloods is now in place. Results from electronic detection systems situated in biochemistry labs are sent to primary care, making changes in serum creatinine concentration easier to spot. This document gives guidance for the outpatient management of AKI and further information is available at: <https://www.thinkkidneys.nhs.uk/resources/primary-care>
- **Improvements are required at discharge from hospital** so that patients who have recovered from AKI have clear plans for follow up and for reintroduction of long term medications that may have been stopped during admission.

How to recognise AKI

The presence of AKI is determined using internationally recognised criteria that are based on individualised changes in serum creatinine concentration with respect to that person's usual (or baseline) value, and reduction in urine volume. **In practice, the urine output criteria can only be applied to hospitalised patients who are catheterised.**

AKI is defined by any of the following:

- Increase in serum creatinine by ≥ 27 micromol/L within 48 hours; or
- Increase in serum creatinine by ≥ 1.5 times baseline, which is known **or presumed to have occurred*** within the prior seven days; or
- Urine volume < 0.5 mL/kg/h for six hours

*** This is crucial, because it is common to see patients with an increase in serum creatinine and longer gap between current value and baseline. In this situation, there are two things to consider:**

- 1. Is the patient acutely unwell?** If so, AKI is more likely.
- 2. Repeating the creatinine within 48-72hrs.** A repeat creatinine will help to determine whether the changes are dynamic or are stable (i.e. more consistent with CKD).

The severity of AKI is described by categorising into three stages (stage 1 being the least severe and stage 3 being the most severe) as follows:

AKI stage	Serum creatinine criteria	Urine output criteria
1	Increase in serum creatinine to 1.5 to 1.9 times baseline; or Increase in serum creatinine by ≥ 27 micromol/L with 48hrs	< 0.5 mL/kg per hour for 6 to 12 hours
2	Increase in serum creatinine to 2.0 to 2.9 times baseline	< 0.5 mL/kg per hour for ≥ 12 hours
3	Increase in serum creatinine to 3.0 times baseline; or Increase in serum creatinine to ≥ 354 micromol/L; or Initiation of kidney replacement therapy; or In patients < 18 yrs, decrease in eGFR to < 35 mL/min/1.73m²	< 0.3 mL/kg per hour for ≥ 24 hours; or anuria for ≥ 12 hours

What causes AKI?

There are many causes of AKI. However, most cases occur in conjunction with co-existing acute illness and are a result of sepsis, hypovolaemia, hypotension or medication effects; these causes, often in combination, account for up to **80% of cases**. This scenario is commonly seen in patients with long term conditions or the frail/elderly.

5-10% of AKI is due to post kidney obstruction, e.g. bladder outflow obstruction.

Intrinsic kidney diseases are less common, but are important not to miss because it is crucial that specialised management of these cases is accessed early. This category includes a variety of less common conditions such as: drug induced tubulo-interstitial nephritis, vasculitis/rapidly progressive glomerulonephritis, myeloma. There are some 'red flag' signs that help to identify this group of AKI patients so they can be referred to nephrology early:

- AKI with blood and protein ($\geq 2+$) on urinalysis
- AKI with systemic symptoms of inflammatory process: vasculitic rash, arthralgia, epistaxis or haemoptysis
- AKI in relation to the introduction of a new drug (PPI, NSAID, antibiotic, diuretic, allopurinol) without any other explanations for AKI
- AKI and high calcium

Who is at risk of developing AKI?

The following are risk factors for the development of AKI:

- | | |
|---|---|
| <ul style="list-style-type: none">• Patient specific<ul style="list-style-type: none">○ Increasing age○ CKD○ Diabetes mellitus○ Heart failure○ Liver disease | <ul style="list-style-type: none">• Situation specific<ul style="list-style-type: none">○ Hypovolaemia, dehydration, reduced oral intake○ Hypotension○ Sepsis○ Post-operative |
|---|---|

In addition, medications also impact on this. Some, such as NSAIDs, are nephrotoxic and will directly increase the risk of AKI. Diuretics may worsen hypovolaemia. ACE inhibitors (ACEi) and angiotensin receptor blockers (ARB) reduce the ability of the kidney to adapt to changes in perfusion pressure. One of the actions of ACEi and ARBs that account for their reno-protective effects in diabetic nephropathy and proteinuric CKD is the reduction in efferent glomerular arteriolar tone. However this action also reduces the ability to maintain glomerular filtration pressure in the face of dehydration/hypoperfusion.

SGLT2 inhibitors cause afferent arteriole vasoconstriction and reduction in hyperfiltration and can cause transient acute reduction of glomerular filtration rate, followed by a progressive recovery and stabilization of kidney function. This is not harmful (i.e. not associated with AKI or

with longer term eGFR loss) and there is some evidence that SGLT2i may even reduce the risk of AKI.

Initiation of ACEi/ARB or SGLT2 inhibitors

Current guidance is that creatinine should be checked one week after initiation of ACEi/ARB and that an increase of up to 20% is acceptable, as long as this rise is stable. This rise reflects the changes in glomerular haemodynamics as above and is **not** a sign of nephrotoxicity. AKI would only be diagnosed if this rise was greater than 50% (the increment of $>27\mu\text{mol/l}$ does not apply because the gap between blood tests should be $>48\text{hrs}$).

Routine checking of biochemistry in the weeks following initiation of SGLT2i is not necessary.

How to reduce the risk of AKI

This can be approached in different ways. Some interventions may be undertaken on a systematic, practice wide basis. Others may be more appropriate for individual patient management, supported by the correct tools and information.

1. Sick day rules for high risk patient groups

This strategy is endorsed by the NICE AKI clinical guideline. Patients at increased risk of AKI are advised to temporarily suspend certain medications during periods of acute illness, particularly when at risk of dehydration. An example leaflet, which has been formulated with patient and professional input, is included as an appendix. Sick day rules for AKI usually refer to ACEi/ARB, which can be stopped temporarily without risk (including in heart failure). Diuretics may sometimes be included in this advice depending on the clinical scenario (greater care needed not to precipitate decompensation in patients with moderate to severe heart failure).

There are different options to disseminate sick day rule advice to patients. It is unlikely that one model will suit all GP practices. The following are some examples of possible models, none of these are meant to be prescriptive:

- Giving out leaflets during consultations on an ad hoc basis
- IT search to identify patients with chronic conditions and prescribed an ACEi/ARB. This resulting list may allow systematic distribution of leaflets (this doesn't specify who at the practice disseminates the leaflets, or how this is done) or be used to trigger automatic reminders
- Adding leaflets to repeat prescriptions for patients on ACEi/ARB

In time, resources that will help support these strategies are likely to be developed locally and nationally that will be shared.

2. When a patient at increased risk of AKI presents with an intercurrent illness, check kidney function

There is no point in checking blood tests if the patient you are seeing requires immediate admission to hospital on clinical grounds. However, checking kidney function in unwell patients with risk factors for AKI may allow earlier detection and intervention.

3. Avoid prescription of long term NSAIDs where possible, particularly in high risk patients and those with CKD

4. Avoid prescribing triple combination of spironolactone, NSAID and ACEi/ARB

Evidence exists that this triple combination confers an increased risk of AKI in its own right. Equally, joint prescription of ACEi and ARBs is now generally not recommended (MHRA safety notice) and the combination of spironolactone and ACEi/ARB in CKD patients should also be carefully considered.

5. Monitor kidney function one week after the introduction of the following medications:

- ACEi/ARB
- Spironolactone/furosemide
- Trimethoprim in patients with CKD
- Loop diuretics in patients with CKD

How to manage a patient with AKI detected in primary care

Not all patients with a rise in creatinine will need admission to hospital. The following is a guide but clinical judgement must always prevail.

1. Is this definitely AKI?

As detailed above, not every patient with a rise in creatinine will have a recent baseline to compare with. The result has to be taken in clinical context, and a repeat creatinine (after 48-72hrs) if the situation allows may help distinguish dynamic changes in serum creatinine from a more stable CKD picture.

2. Is the patient acutely unwell?

A blood test has to be interpreted within the overall clinical scenario, and this is especially important for AKI. Blood tests suggesting AKI in an unwell patient (especially if there are signs of sepsis) should increase clinical concern.

3. How severe is the AKI?

Increasing severity of AKI correlates with higher risk of worse outcomes. AKI stage 3 should be managed in secondary care.

4. Think about cause of AKI: does the patient have any red flag signs for urinary obstruction or intrinsic kidney disease?

DIP THE URINE: this is an important diagnostic step.

- AKI and negative urinalysis: usually pre-kidney causes (also consider drug causes)
- AKI with $\geq 2+$ blood and protein: wider differential diagnosis that includes intrinsic kidney disease

If there are clinical pointers to urinary obstruction or intrinsic kidney disease, these patients will need specialist referral (obstruction to urology, intrinsic kidney disease to nephrology)

5. Approach to outpatient management of AKI

- Avoid or correct dehydration
- Medication review

- i. Consider temporary suspension of ACEi/ARB +/- diuretics
 - ii. Consider temporary suspension of metformin (to avoid risk of lactic acidosis)
 - iii. Stop nephrotoxic medications such as NSAIDs
 - iv. In the absence of an obvious cause of AKI, consider if any new drugs have been introduced that have a temporal relationship to the change in kidney function: especially antibiotics and PPIs
- Early review and repeat U/Es: seek help from nephrology SpR on call (available via RDH/CRH switchboard) for patients who are getting worse despite the above

6. If in doubt, discuss with nephrology

- Contact nephrology SpR on call (via RDH/CRH switchboard)

Appendix: AKI sick day rules leaflet

Please open the attachment on the shared care pathology page

Other Useful Resources

- UK Kidney Association
[UK Kidney Association | The leading professional body for the UK kidney community](#)
This link includes many resources for healthcare professionals as well as patients.
- AKI Toolkit for GP's
<https://www.rcgp.org.uk/aki>
- NICE AKI guidelines (NG148) last update Oct 2024
<https://www.nice.org.uk/guidance/ng148>

Document Controls

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